Math Thermo
class # 08
02/12/2026

Free Energy Density

Suppose that $V > 0$ is fixed. Define, as before,

$$\rho := \frac{N}{V}, \qquad \text{mole number density}$$

Then, since, for any $\lambda > 0$,

$$\lambda F(T, V, N) = F(T, \lambda V, \lambda N),$$

it follows that

$$\frac{1}{V} F(T, V, N) = F\left(T, \frac{V}{V}, \frac{N}{V}\right)$$
$$= F(T, 1, \rho).$$

Now, define the free energy density

$$f = f(T, \rho) := F(T, 1, \rho).$$

It then follows that

$$\boxed{F(T, V, N) = V f(T, \rho).}$$

Recall, that Euler's Equation in density form is

(8.1) $$\tilde{u}(s, \rho) = T_0(s, \rho)\, s - P_0(s, \rho) + \mu_0(s, \rho)\, \rho .$$

Thus,

$$F(T,V,N) = \tilde{U}(S_T, V, N) - T S_T$$

$$V f(T,\rho) = V u\left(\frac{S_T}{V}, \rho\right) - V T \frac{S_T}{V}$$

$$f(T,\rho) = u(s_T, \rho) - T s_T,$$

where

$$s_T := \frac{S_T}{V}$$

and $S_T = S_T(V,N)$ is the unique solution to

$$T_U(S_T(V,N), V, N) = T.$$

Since T is homogeneous of degree zero,

$$T = T_U(S_T(V,N), V, N)$$

$$= T_U(s_T, 1, \rho)$$

$$= T_U(s_T, \rho)$$

Thus, we have proven the following:

<u>Theorem</u> (8.1): The free energy density satisfies

$$f(T,\rho) = u(s_T, \rho) - T s_T,$$

where $s_T = s_T(\rho)$ is the unique solution to

$$T_U(s_T(\rho), \rho) = T.$$

Theorem (8.2): We have

$$dF = -S\,dT - P\,dV + \sum_{j=1}^{r} \mu_j \, dN_j$$

by which we mean $F = F(T, V, \vec{N})$ and

$$\frac{\partial F}{\partial T}(T, V, N) = -S_T(V, \vec{N}),$$

$$\frac{\partial F}{\partial V}(T, V, N) = -P_U(S_T(V, \vec{N}), V, \vec{N}),$$

$$\frac{\partial F}{\partial N_j}(T, V, N) = \mu_{U,j}(S_T(V, \vec{N}), V, \vec{N}),$$

where $S_T = S_T(V, \vec{N})$ is the unique solution of the equation

$$T_U(S_T(V, \vec{N}), V, \vec{N}) = T.$$

Proof: Exercise. ///

Example (8.3): Suppose that

$$\tilde{S}(U, V, N) = \left(\frac{N V U R^2}{v_0 \theta} \right)^{1/3},$$

where $R, v_0, \theta > 0$ are constants.

Then,

$$\tilde{U}(S,V,N) = \frac{v_0 \theta S^3}{NVR^2},$$

as we have observed many times, and

$$T_U(S,V,N) = \frac{\partial \tilde{U}}{\partial S}$$

$$= \frac{3 v_0 \theta S^2}{NVR^2}.$$

It follows that, for any $T \geq 0$

$$S_T(V,N) = \left(\frac{NTVR^2}{3 v_0 \theta}\right)^{1/2}.$$

Therefore,

$$F(T,V,N) = \tilde{U}(S_T(V,N),V,N) - TS_T(V,N)$$

$$= \frac{v_0 \theta (S_T(V,N))^3}{NVR^2} - T\, S_T(V,N)$$

$$= \frac{v_0 \theta}{NVR^2}\left(\frac{NTVR^2}{3 v_0 \theta}\right)^{3/2}$$

$$- T\left(\frac{NTVR^2}{3 v_0 \theta}\right)^{1/2}$$

$$= \left(\frac{NTVR^2}{3v_0\theta}\right)^{1/2} \left\{ \frac{\cancel{v_0}\,\cancel{\theta}\,\cancel{N}T\cancel{V}\cancel{R^2}}{\cancel{N}\cancel{V}\cancel{R^2}\,3\,\cancel{v_0}\,\cancel{\theta}} - T \right\}$$

$$= -\frac{2}{3}\left(\frac{NVR^2}{3v_0\theta}\right)^{1/2} T^{3/2}.$$

The partial with respect to T is

$$\frac{\partial F}{\partial T} = -\left(\frac{NTVR^2}{3v_0\theta}\right)^{1/2}$$

$$= -S_T(V,N).$$

The pressure is given by

$$P_U(S,V,N) = \frac{-\partial \tilde{U}}{\partial V} = \frac{v_0\theta S^3}{NV^2R^2}.$$

Therefore,

$$P_U(S_T(V,N),V,N) = \frac{v_0\theta}{NV^2R^2}\left(\frac{NTVR^2}{3v_0\theta}\right)^{3/2}$$

$$= \left(\frac{T}{3}\right)^{3/2}\left(\frac{NR^2}{v_0V\theta}\right)^{1/2}.$$

The partial of F with respect to V is

$$\frac{\partial F}{\partial V} = -\frac{1}{3}\left(\frac{NR^2}{3Vv_0\theta}\right)^{1/2} T^{3/2}$$

$$= -\left(\frac{T}{3}\right)^{3/2}\left(\frac{NR^2}{v_0 V\Theta}\right)^{1/2}$$

$$= -P_V(s_T(V,N),V,N).$$

The reader should confirm that

$$\frac{\partial F}{\partial N_j} = \mu_{V,j}(s_T(V,N),V,N).$$

<u>Theorem</u> (8.4): We have

$$df = -s\,dT + \sum_{j=1}^{r} \mu_j \, d\rho_j$$

by which we mean $f = f(T,\rho)$ and

$$\frac{\partial f}{\partial T}(T,\vec{\rho}) = -s_T(\vec{\rho}),$$

$$\frac{\partial f}{\partial \rho_j}(T,\vec{\rho}) = \mu_{V,j}(s_T(\vec{\rho}),\vec{\rho}),$$

where $s_T(\vec{\rho})$ is the unique solution to

$$T_V(s_T(\vec{\rho}),\vec{\rho}) = T.$$

<u>Proof</u>: Exercise. ///

A good example goes a long way.

Example (8.5): We will use our standard model. Suppose that

$$\tilde{U}(S,V,N) = \frac{v_0 \theta S^3}{NVR^2}.$$

Then

$$u(s,\rho) = \frac{1}{V}\tilde{U}(S,V,N)$$
$$= \tilde{U}(s,1,\rho)$$
$$= \frac{v_0 \theta s^3}{\rho R^2}$$

and

$$T_U(s,\rho) = \frac{3 v_0 \theta s^2}{\rho R^2}.$$

Thus,

$$s_T(\rho) = \left(\frac{\rho R^2 T}{3 v_0 \theta}\right)^{1/2},$$

and

$$f(T,\rho) = u(s_T(\rho),\rho) - T s_T(\rho)$$

$$= \frac{v_0 \theta (s_T(\rho))^3}{\rho R^2} - T\left(\frac{\rho R^2 T}{3 v_0 \theta}\right)^{1/2}$$

$$= \frac{v_0 \theta}{\rho R^2} \left(\frac{\rho R^2 T}{3 v_0 \theta} \right)^{3/2} - T \left(\frac{\rho R^2 T}{3 v_0 \theta} \right)^{1/2}$$

$$= \left(\frac{\rho R^2 T}{3 v_0 \theta} \right)^{1/2} \left\{ \frac{\cancel{v_0} \cancel{\theta}}{\cancel{\rho} \cancel{R^2}} \frac{\cancel{\rho} \cancel{R^2} T}{3 \cancel{v_0} \cancel{\theta}} - T \right\}$$

$$= -\frac{2 T^{3/2}}{3} \left(\frac{\rho R^2}{3 v_0 \theta} \right)^{1/2}.$$

The partial with respect to T is

$$\frac{\partial f}{\partial T}(T, \rho) = -\left(\frac{\rho T R^2}{3 v_0 \theta} \right)^{1/2}$$

$$= -s_T(\rho).$$

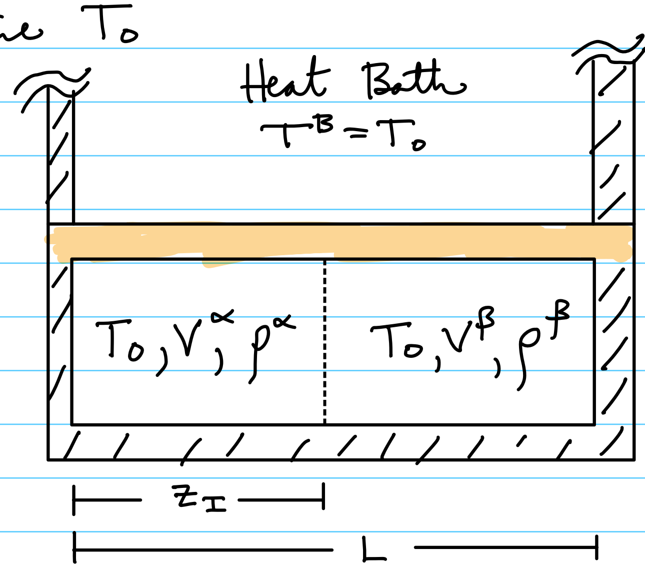
Maxwell's Common Tangent Construction

Here we will review a situation for which the Helmholtz potential is important.

Consider a unary, isothermal, two phase system, for example liquid and solid iron.

We will assume the β phase is the high-density phase (a solid perhaps) and the α phase is the low-density phase (a liquid, perhaps).

Both phases are in contact with a heat bath at temperature T_0



Here the densities, ρ^α and ρ^β, play the roles of the numbers of moles, N^α and N^β.

We will assume, for simplicity, that parameters vary in only one spatial dimension, z.

Suppose the cross-section area of the container is A. Then

$$V^\alpha = A\, z_I$$

$$V^\beta = A\,(L - z_I)$$

$$V^\alpha + V^\beta = A \cdot L =: V_0 \qquad \left(\begin{array}{c}\text{total volume}\\ \text{constant}\end{array}\right)$$

Now,

$$N^\alpha = V^\alpha \rho^\alpha,$$

$$N^\beta = V^\beta \rho^\beta.$$

Here

$$[\rho^\alpha] = \text{moles / unit vol}$$

$$N_0 = N^\alpha + N^\beta \qquad \left(\begin{array}{c}\text{total mole}\\ \text{number const}\end{array}\right)$$

$$= V^\alpha \rho^\alpha + V^\beta \rho^\beta$$

$$= A z_I \rho^\alpha + A(L - z_i)\rho^\beta$$

It will be convenient to introduce

$$\rho_0 = \frac{N_0}{V_0} = \frac{N_0}{L \cdot A}.$$

Hence,

(8.2) $$\boxed{\rho_0 \cdot L = z_I \rho^\alpha + (L - z_I)\rho^\beta.}$$

Assume that

$$F_{\alpha+\beta}(V^\alpha, V^\beta, \rho^\alpha, \rho^\beta) = V^\alpha f^\alpha(\rho^\alpha) + V^\beta f^\beta(\rho^\beta).$$

We suppress the temperature dependence, for simplicity.
The variables f^α and f^β are called free energy densities. We can eliminate the volume variables using constraints:

$$F_{\alpha+\beta}(V^\alpha, V^\beta, \rho^\alpha, \rho^\beta) = V^\alpha f^\alpha(\rho^\alpha) + V^\beta f^\beta(\rho^\beta)$$

$$= A z_I f^\alpha(\rho^\alpha) + A(L - z_I) f^\beta(\rho^\beta).$$

Using the mass constraint (8.2), we have

$$z_I = \frac{L(\rho_0 - \rho_\beta)}{\rho^\alpha - \rho^\beta}$$

or

(8.3) $$\boxed{z_I = \frac{L(\rho^\beta - \rho_0)}{\rho^\beta - \rho^\alpha}}$$

Similarly, we have

(8.4) $$\boxed{L - z_I = \frac{L(\rho_0 - \rho^\alpha)}{\rho^\beta - \rho^\alpha}}$$

Thus,

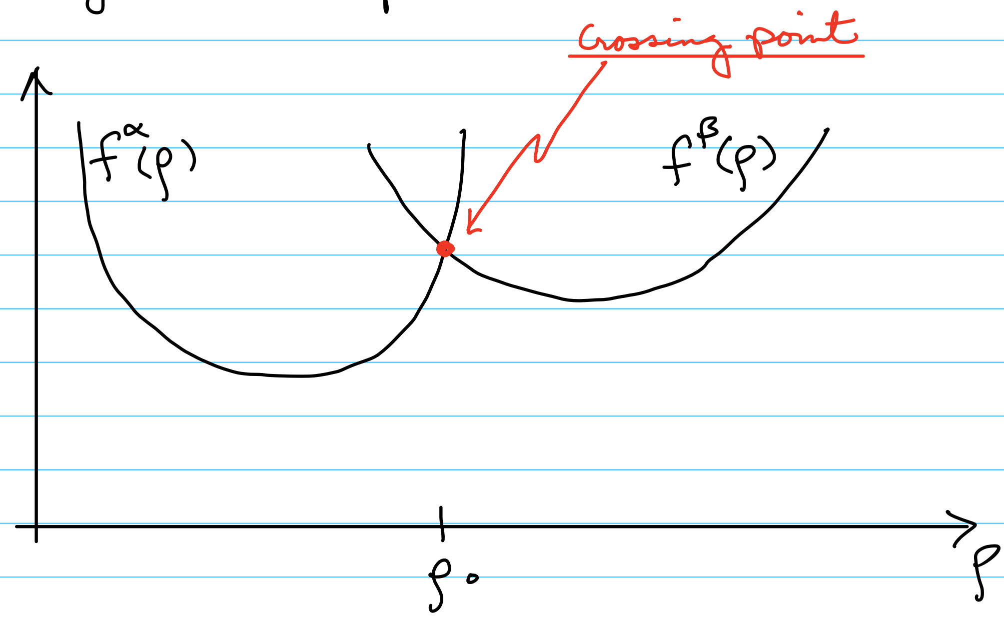
$$F_{\alpha+\beta}(V^\alpha, V^\beta, \rho^\alpha, \rho^\beta) = \overbrace{L \cdot A}^{=V_0} \cdot \tilde{f}(\rho^\alpha, \rho^\beta),$$

where

$$\tilde{f}(\rho^\alpha, \rho^\beta) = \frac{\rho^\beta - \rho_0}{\rho^\beta - \rho^\alpha} f^\alpha(\rho^\alpha) + \frac{\rho_0 - \rho^\alpha}{\rho^\beta - \rho^\alpha} f^\beta(\rho^\beta).$$

Recall, the setting is isothermal, $T^\alpha = T^\beta = T_0$. So equilibrium is characterized by a minimum in the free energy. Thus, we seek a minimizer of $\tilde{f}$.

The setting is as follows:



We want ρ^α and ρ^β that minimize $\tilde{F}$.

$$0 = \frac{\partial \tilde{f}}{\partial \rho^\alpha}(\rho^\alpha, \rho^\beta)$$

$$0 = \frac{\partial \tilde{f}}{\partial \rho^\beta}(\rho^\alpha, \rho^\beta)$$

Now,

$$\frac{\partial \tilde{f}}{\partial \rho^\alpha} = \frac{\rho^\beta - \rho_0}{(\rho^\beta - \rho^\alpha)^2} f^\alpha(\rho^\alpha) + \frac{\rho^\beta - \rho_0}{\rho^\beta - \rho^\alpha} f^{\alpha\prime}(\rho^\alpha)$$

$$+ \frac{\rho_0 - \rho^\beta}{(\rho^\beta - \rho^\alpha)^2} f^\beta(\rho^\beta)$$

$$= \frac{\rho^\beta - \rho_0}{\rho^\beta - \rho^\alpha} \left\{ \frac{f^\alpha(\rho^\alpha) - f^\beta(\rho^\beta)}{\rho^\beta - \rho^\alpha} + f^{\alpha\prime}(\rho^\alpha) \right\}$$

and

$$\frac{\partial \tilde{f}}{\partial \rho^\beta} = \frac{\rho_0 - \rho^\alpha}{(\rho^\beta - \rho^\alpha)^2} f^\alpha(\rho^\alpha)$$

$$- \frac{\rho_0 - \rho^\alpha}{(\rho^\beta - \rho^\alpha)^2} f^\beta(\rho^\beta) + \frac{\rho_0 - \rho^\alpha}{\rho^\beta - \rho^\alpha} f^{\beta\prime}(\rho^\beta)$$

$$= \frac{\rho_0 - \rho^\alpha}{\rho^\beta - \rho^\alpha} \left\{ \frac{f^\alpha(\rho^\alpha) - f^\beta(\rho^\beta)}{\rho^\beta - \rho^\alpha} + f^{\beta\prime}(\rho^\beta) \right\}$$

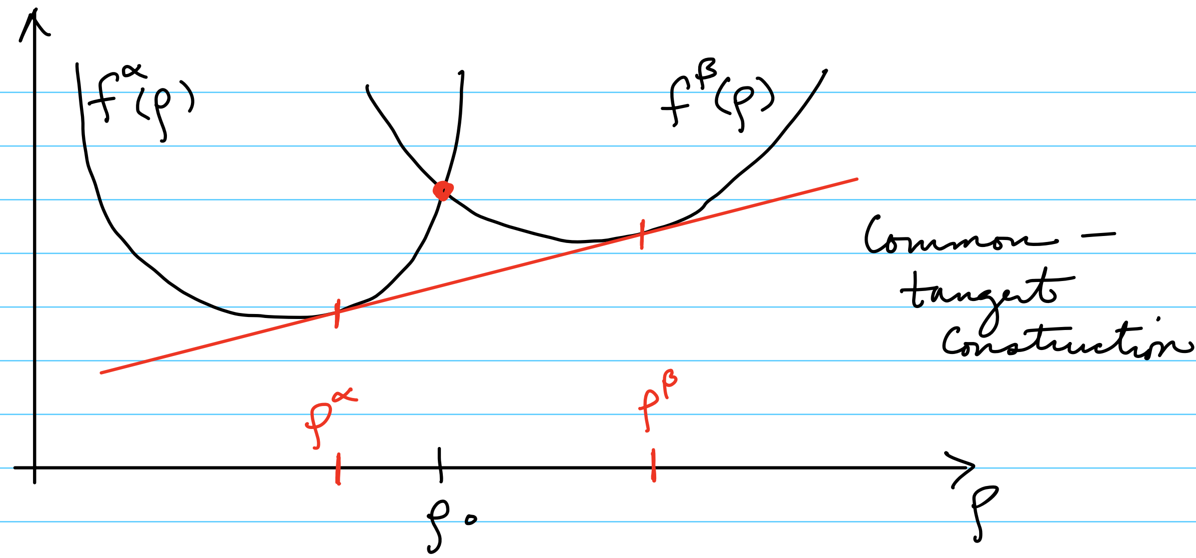
Thus, the minimizer must satisfy

(8.5)
$$\boxed{\frac{f^\beta(\rho^\beta) - f^\alpha(\rho^\alpha)}{\rho^\beta - \rho^\alpha} = f^{\alpha\prime}(\rho^\alpha)}$$

and

(8.6)
$$\boxed{\frac{f^\beta(\rho^\beta) - f^\alpha(\rho^\alpha)}{\rho^\beta - \rho^\alpha} = f^{\beta\prime}(\rho^\beta)}$$

Graphically, these solutions represent the following scenario:



For everything to make sense, we require that

$$\rho^\alpha < \underbrace{\rho_0}_{\text{average density}} < \rho^\beta$$

ρ^α: low density phase; ρ^β: high density phase

What happens if, say,

$$\rho^\alpha < \rho^\beta < \rho_0,$$

that is, the average is bigger than the density of the high-density phase, β?

Recall,

$$z_I = \frac{L(\rho^\beta - \rho_0)}{\rho^\beta - \rho^\alpha}.$$

If $\rho^\alpha < \rho^\beta < \rho_0$, it follows that

$$z_I < 0,$$

which is unphysical.

On the other hand if $\rho_0 < \rho^\alpha < \rho^\beta$, then

$$z_I > L,$$

which is unphysical.

But, if

$$\rho^\alpha < \rho_0 < \rho^\beta,$$

it follows that z_I is fully determined and

$$0 < z_I < L,$$

as desired.

Recall that the chemical potentials satisfy

$$\mu^\alpha = \frac{\partial f^\alpha}{\partial \rho^\alpha}$$

$$\mu^\beta = \frac{\partial f^\beta}{\partial \rho^\beta}$$

So that equilibrium occurs when

$$\mu^\alpha = \mu^\beta.$$

Unary, Isothermal Phase Diagrams
(at constant volume)

Let us consider the following phase diagram